

Cybersecurity Research Report
Network Security & Zero Trust Network Access (ZTNA)
with Tailscale

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1. Introduction

Cybersecurity is the practice of protecting systems, networks, and data from unauthorized access, attacks, and damage. With the rapid growth of the internet and digital technologies, cybersecurity has become a critical requirement in both personal and professional environments. Organizations today rely heavily on secure systems to store and process sensitive information.

Network security is a key part of cybersecurity that focuses on protecting communication between devices. It ensures that data is transmitted safely without being intercepted or altered by attackers. As modern systems become more connected, traditional security approaches are no longer sufficient, which has led to the adoption of advanced models like Zero Trust. In my opinion, traditional security approaches are becoming less effective due to increasing remote access and evolving cyber threats.

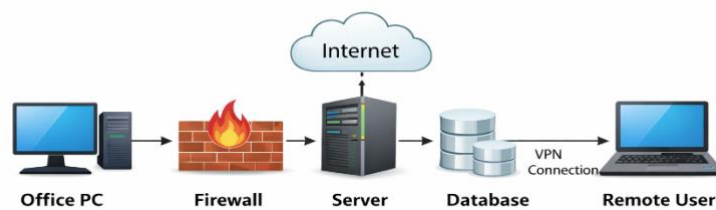


Figure 1: Basic Network Communication Model

2. Problem Statement

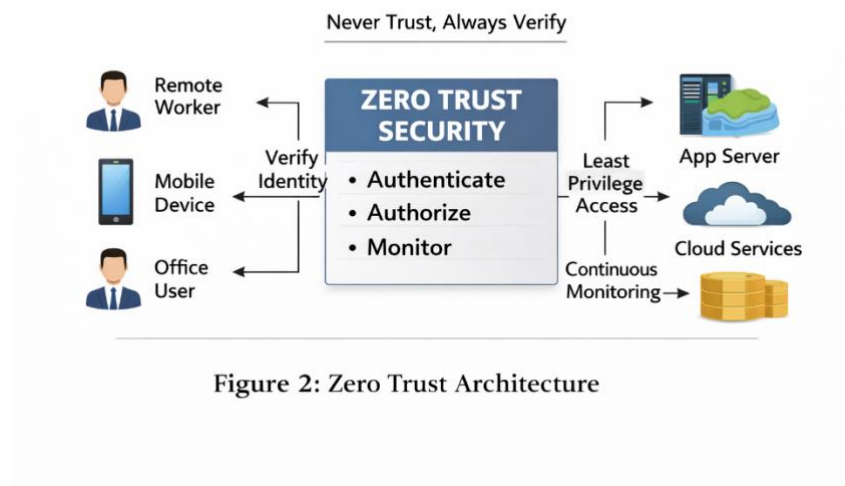
Traditional network security models such as Virtual Private Networks (VPNs) operate on a trust-based approach. Once a user is authenticated, they are often granted broad access to the entire network. This creates significant security risks, especially if credentials are compromised by attackers.

These models also lack continuous monitoring and verification, making it difficult to detect suspicious activity. With the rise of remote work and cloud computing, these vulnerabilities have become more serious. Attackers can exploit these weaknesses to gain unauthorized access to sensitive systems and data. This highlights the need for a more secure and controlled access model like Zero Trust.

3. Zero Trust Network Access (ZTNA)

Zero Trust Network Access (ZTNA) is a modern security model based on the principle 'Never trust, always verify'. Unlike traditional VPNs, which provide broad access after authentication, ZTNA restricts access to specific resources, reducing the risk of unauthorized access and minimizing the attack surface.

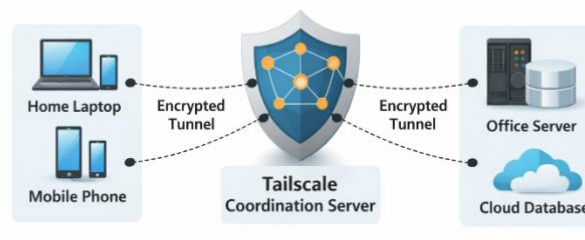
ZTNA uses identity-based access control, ensuring that only authorized users can access specific resources. It also follows the principle of least privilege, where users are given only the minimum level of access required. Continuous monitoring further enhances security by detecting unusual behavior in real time.



4. Case Study: Tailscale

Tailscale is a modern networking solution that implements Zero Trust principles to provide secure communication between devices. It creates a private network where devices can connect safely without exposing them to the public internet. This reduces the risk of external attacks.

Tailscale uses identity-based authentication and encryption to ensure secure data transmission. It simplifies network configuration while maintaining strong security. This makes it a practical example of how ZTNA can be applied in real-world scenarios.

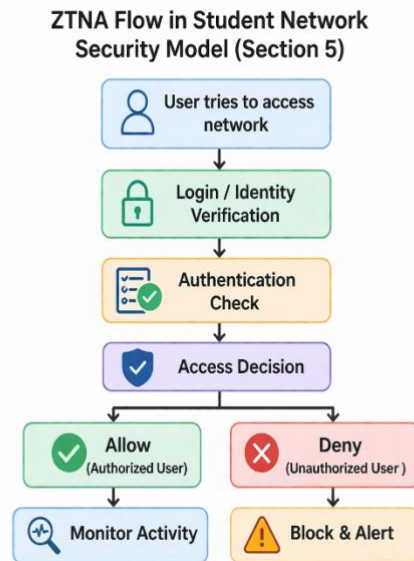


5. Proposed Idea: Student Network Security Model

The proposed idea is to create a beginner-level project that demonstrates ZTNA concepts in a simple network environment. Using Cisco Packet Tracer, a small network can be designed with multiple devices communicating with each other.

Access control can be applied to ensure that only authorized users can interact within the network. Wireshark can be used to capture and analyze network traffic, helping identify normal and suspicious patterns. This project provides practical exposure to network security concepts. This model can be further improved by implementing

role-based access control, where different users such as students and administrators have different levels of permissions, similar to real-world systems.



The flowchart represents the step-by-step process of secure access, starting from user authentication to continuous monitoring and access control.

6. Conclusion

Cybersecurity is evolving rapidly due to increasing digital threats. Traditional security models are no longer sufficient to protect modern systems. Zero Trust Network Access provides a more secure and efficient approach by ensuring strict verification and controlled access.

This research highlights the importance of network security and demonstrates how ZTNA can improve system protection. Through this study, a strong foundation has been developed in understanding modern security approaches and their practical applications.

Through this research, I have developed a clear understanding of modern cybersecurity approaches and aim to explore and implement these concepts further.